Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER B.TECH DEGREE EXAMINATION, OCTOBER 2021

Course Code: EST130

Course Name: BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING

PART I: BASIC ELECTRICAL ENGINEERING

PART A

Answer all questions. Each question carries 4 marks

Q1 A conductor of length 0.5m kept at right angles to a uniform magnetic field of flux density $2 Wb/m^2$ moves with a velocity of 75 m/s at an angle of 60^0 to the field. Calculate the emf induced in the conductor.

Ans EMF induced, $e = Blv \sin\theta = 2\times 0.5\times 75\times \sin 60 = 64.95V$

Q2 Define mutual inductance. Two coupled coils of self inductance 0.8H and 0.35H have a coefficient of coupling 0.9. Find the mutual inductance between the coils.

Ans Mutual Inductance between the two coils is defined as the property of the coil due to which it opposes the change of current in the neighbouring coil.

Coefficient of Coupling, $k = \dfrac{M}{\sqrt{L_1 L_2}}$

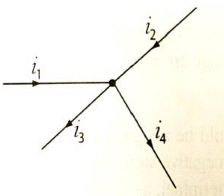
Mutual Inductance,

$$M = k\sqrt{L_1 L_2} = 0.9\times\sqrt{0.8\times 0.35} = 0.476H$$

Q3 State and explain Kirchhoff's laws with examples.

Ans **Kirchhoff's Current Law (KCL):** It states that at any instant of time, the algebraic sum of currents at a node is zero.

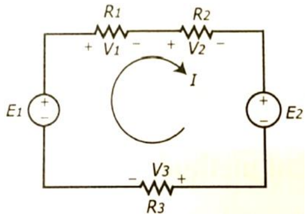
Consider the node as shown below. I_1, i_2, i_3 and i_4 are currents at the node.



According to KCL,

$$i_1 + i_2 - i_3 - i_4 = 0$$
$$i_1 + i_2 = i_3 + i_4$$

Kirchhoff's Voltage Law (KVL): It states that the algebraic some of voltages around a closed path at any instant of time is zero.

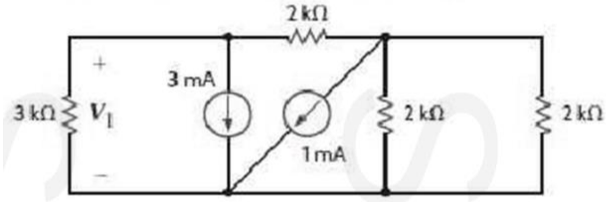
		Consider the closed path as shown below. According to KVL, $E_1 - V_1 - V_2 - E_2 - V_3 = 0$ 
	Q4	Find the trignometrical, exponential and polar forms of the vector 8+j6.
	Ans	$v = 8 + j6$ Magnitude $V = \sqrt{8^2 + 6^2} = 10V$ Phase $\phi = \tan^{-1}\left(\frac{6}{8}\right) = 36.87°$ Trignometrical representation = $V(\cos\phi + j\sin\phi) = 10(\cos 36.87° + j\sin 36.87°)$ Exponential representation = $Ve^{j\phi} = 10e^{j36.87}$ Polar representation = $V\angle\phi = 10\angle 36.87°$
	Q5	Define (i) active power, (ii) reactive power, (iii) apparent power and (iv) power factor of an ac circuit.
	Ans	i) Active Power: It is the actual power dissipated in the circuit. Its unit is Watt (W). $P = VI\cos\phi$ ii) Reactive Power: It is the power associated with reactive components of the circuit. It is not dissipated in the circuit, but stored and released during each AC cycle. It is also called wattles component of power. Its unit is Volt-Ampere Reactive (VAR). $Q = VI\sin\phi$ iii) Apparent Power: The product of rms values of voltage and current in an AC circuit is called apparent power. Its unit is Volt-Ampere (VA). $S = VI$ iv) Power factor: It is the ratio of active power to apparent power. In other words, it is the cosine of the angle between voltage and ciurrent. $pf = \frac{Active\ Power}{Apparent\ Power} = \cos\phi$

PART B

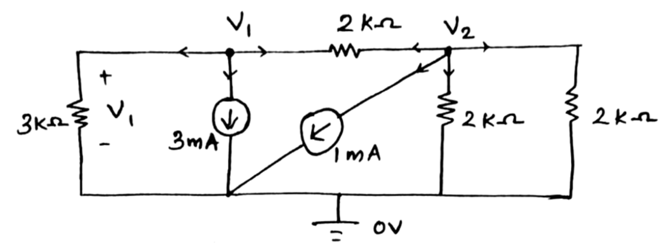
Answer any one full question from each module. Each question carries 10 marks

Module 1

Q6 Use nodal analysis to find V_1 in the given circuit.



Ans



<u>At node - 1</u>

$$\frac{V_1}{3} + 3 + \frac{V_1 - V_2}{2} = 0$$

$$V_1\left[\frac{1}{3} + \frac{1}{2}\right] - \frac{V_2}{2} = -3 \quad \text{——①}$$

<u>At node - 2</u>

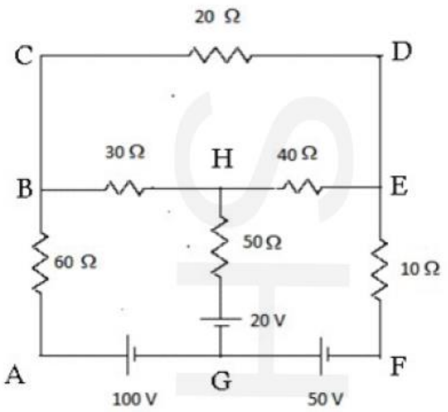
$$\frac{V_2 - V_1}{2} + 1 + \frac{V_2}{2} + \frac{V_2}{2} = 0$$

$$-\frac{V_1}{2} + V_2\left[\frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right] = -1 \quad \text{——②}$$

Solving ① and ②, we get

$$V_1 = \underline{\underline{-5\ V}}$$

$$V_2 = \underline{\underline{-2.33\ V}}$$

<table>
<tr><td></td><td>Q7</td><td>Find the current in each branch of the following circuit using mesh analysis?

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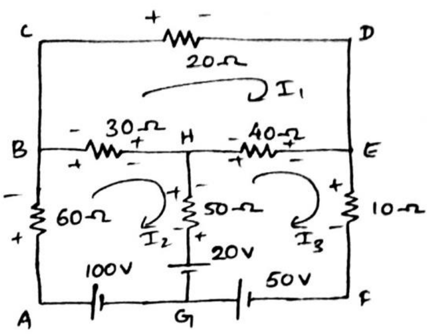
<tr><td></td><td>Ans</td><td>

Mesh - 1

$-20I_1 - 40I_1 + 40I_3 - 30I_1 + 30I_2 = 0$

$-90I_1 + 30I_2 + 40I_3 = 0$ —— 1

Mesh - 2

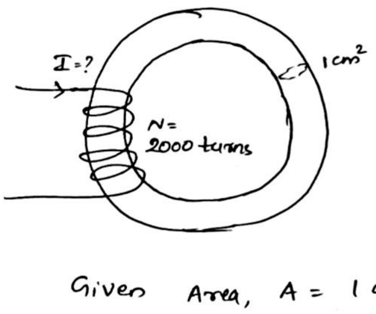
$100 - 60I_2 - 30I_2 + 30I_1 - 50I_2 + 50I_3 - 20 = 0$

$-30I_1 + 140I_2 - 50I_3 = 80$ —— 2

Mesh - 3

$20 - 50I_3 + 50I_2 - 40I_3 + 40I_1 - 10I_3 + 50 = 0$

$-40I_1 - 50I_2 + 100I_3 = 70$ —— 3</td></tr>
</table>

		solving 1, 2 and 3, $I_1 = 1.493\ A$ $I_2 = 1.65\ A$ $I_3 = 2.12\ A$ current in various braches are shown below current in $GAB = I_2 = 1.65\ A$ $BCDE = I_1 = 1.493A$ $BH = I_2 - I_1 = 0.157\ A$ $HG = I_2 - I_3 = -0.47 A$ $HE = I_3 - I_1 = 0.627 A$ $EFG = I_3 = 2.12\ A$
		Module 2
	Q8	An iron ring of cross sectional area 1cm2 is wound with a coil of 2000 turns. Calculate the magnetising current required to produce a flux of 0.1 mWb in the iron path if mean length of the path is 30cm and relative permeability of iron is 2500. Neglect magnetic leakages and fringing.
	Ans	 Given Area, $A = 1\ cm^2 = 1 \times 10^{-4}\ m^2$ length, $l = 30\ cm = 0.3\ m$. $\mu_r = 2500$ $\phi = 0.1\ mWb = 0.1 \times 10^{-3}\ Wb$ $N = 2000$ turns. $I = ?$

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$$\text{Reluctance, } S = \frac{l}{\mu_0 \mu_r A} = \frac{0.3}{4\pi \times 10^{-7} \times 2500 \times 1 \times 10^{-4}}$$

$$= \underline{\underline{954929.66}} \text{ AT/Wb}$$

$$\text{MMF} = \text{Flux} \times \text{reluctance} = \phi \times S$$

$$NI = \phi S$$

$$I = \frac{\phi S}{N} = \frac{0.1 \times 10^{-3} \times 954929.66}{2000}$$

$$= 0.\underline{\underline{04775}} \text{ A}$$

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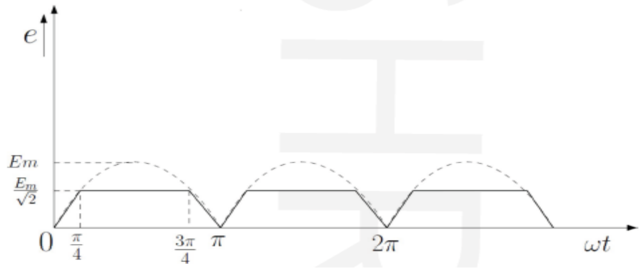
<tr><td></td><td>Q9</td><td>A full wave rectified sine function is clipped at 0.707 of its maximum value as shown in figure. Find the average and rms values of the function.



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<tr><td></td><td>Ans</td><td>Given function can be expressed as,

$$f = \begin{cases} E_m \sin \omega t & 0 \le \omega t < \frac{\pi}{4} \\ E_m/\sqrt{2} & \pi/4 \le \omega t < \frac{3\pi}{4} \\ E_m \sin \omega t & \frac{3\pi}{4} \le \omega t < \pi \end{cases}$$

$$\text{Average value, } f_{avg} = \frac{1}{T}\int_0^T f(t)\, dt$$

$$= \frac{1}{\pi}\int_0^{\pi} f(\omega t)\, d\omega t$$

$$= \frac{1}{\pi}\left[\int_0^{\pi/4} E_m \sin \omega t \, d\omega t + \int_{\pi/4}^{3\pi/4} \frac{E_m}{\sqrt{2}}\, d\omega t + \int_{3\pi/4}^{\pi} E_m \sin \omega t \, d\omega t\right]$$

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$$= \frac{E_m}{\pi}\left[\left(-\cos\omega t\right)_0^{\pi/4} + \left[\frac{\omega t}{\sqrt{2}}\right]_{\pi/4}^{3\pi/4} + \left[-\cos\omega t\right]_{3\pi/4}^{\pi} \right]$$

$$= \frac{E_m}{\pi}\left[\left(-\cos\frac{\pi}{4} - -\cos 0\right) + \left(\frac{3\pi/4 - \pi/4}{\sqrt{2}}\right) + \left(-\cos\pi - -\cos\frac{3\pi}{4}\right) \right]$$

$$= \frac{E_m}{\pi} \times \left[2 - \sqrt{2} + \frac{\pi}{2\sqrt{2}}\right]$$

$$= \underline{\underline{0.54\, E_m}}$$

rms value, $f_{rms} = \sqrt{\frac{1}{T}\int_0^T f(t)^2\, dt}$

$$= \sqrt{\frac{1}{\pi}\int_0^{\pi} f(\omega t)^2\, d\omega t}$$

$$= \sqrt{\frac{1}{\pi}\left\{ \int_0^{\pi/4} E_m^2 \sin^2\omega t\, d\omega t + \int_{\pi/4}^{3\pi/4} \frac{E_m^2}{2} d\omega t + \int_{3\pi/4}^{\pi} E_m^2 \sin^2\omega t\, d\omega t \right\}}$$

$$= \frac{E_m}{\sqrt{\pi}}\left\{ \sqrt{\int_0^{\pi/4} \frac{1-\cos 2\omega t}{2} d\omega t + \int_{\pi/4}^{3\pi/4} \frac{1}{2} d\omega t + \int_{3\pi/4}^{\pi} \frac{1-\cos 2\omega t}{2} d\omega t} \right\}$$

$$= \frac{E_m}{\sqrt{\pi}}\left\{ \sqrt{\frac{1}{2}\left(\omega t - \frac{\sin 2\omega t}{2}\right)_0^{\pi/4} + \frac{1}{2}\left(\omega t\right)_{\pi/4}^{3\pi/4} + \frac{1}{2}\left(\omega t - \frac{\sin 2\omega t}{2}\right)_{3\pi/4}^{\pi}} \right\}$$

$$= \frac{E_m}{\sqrt{2\pi}} \sqrt{\left(\frac{\pi}{4} - 0\right) - \left(\frac{\sin\pi/2}{2} - \frac{\sin 0}{2}\right) + \left(\frac{3\pi}{4} - \frac{\pi}{4}\right) + \left(\pi - \frac{3\pi}{4}\right) - \left(\frac{\sin 2\pi - \sin\frac{3\pi}{2}}{2}\right)}$$

$$= \frac{E_m}{\sqrt{2\pi}} \sqrt{\frac{\pi}{4} - \frac{1}{2} + \frac{\pi}{2} + \frac{\pi}{4} - \frac{1}{2}}$$

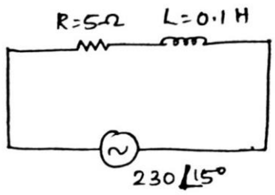
$$= \frac{E_m}{\sqrt{2\pi}} \times \sqrt{\pi - 1}$$

$$= \underline{\underline{0.584\, E_m}}$$

		Module 3
	Q10	A sinusoidal voltage V=230 · 15 of frequency 50 Hz is applied to a series RL circuit consisting of R=5 Ω and =0.1 H. Calculate (i) rms current and its phase angle (ii) power factor (iii) average power (iv) reactive power and (v) apparent power drawn by the circuit.
	Ans	

$V = 230\angle 15^\circ\ V$

$f = 50\ Hz$



Total impedance,

$$Z = R + jX_L$$

$$X_L = \omega L = 2\pi f L = 2\pi \times 50 \times 0.1 = 31.42\ \Omega$$

$$Z = (5 + j\,31.42)\ \Omega = 31.82\angle 80.96^\circ\ \Omega$$

i) Current, $I = \frac{V}{Z} = \underline{7.23\angle -65.96^\circ}\ A$

rms value of current $= \underline{7.23\ A}$

phase angle $= -\underline{65.96^\circ}$

ii) Power factor $= \cos\phi = \frac{R}{Z} = \underline{0.157}$ lag

iii) average power $= VI\cos\phi$

$= 230 \times 7.23 \times 0.157$

$= \underline{261.08}\ W$

iv) reactive power $= VI\sin\phi$

$= 230 \times 7.23 \times 0.988$

$= \underline{1642.24}\ VAR$

v) apparent power $= VI = \underline{1662.9}\ VA$

	Q11	A balanced 3 phase load consists of 3 coils each of resistance 6 Ω and inductive reactance of 8 Ω. Determine the line current and power absorbed when the coils are (i) star connected (ii) delta connected across 400V, 3 phase supply.
	Ans	Given that, $V_L = 400\ V$ $R = 6\Omega$ $X_L = 8\Omega$ $Z = \sqrt{R^2+X_L^2} = \sqrt{6^2+8^2} = \underline{\underline{10\Omega}}$ **Star connected** $V_{Ph} = \frac{400}{\sqrt{3}}\ V$ $I_{Ph} = \frac{V_{Ph}}{Z} = \underline{\underline{23.09}}\ A$ Line current, $I_L = I_{Ph} = \underline{\underline{23.09\ A}}$ power absorbed, $P = \sqrt{3} V_L I_L \cos\phi$ $\cos\phi = \frac{R}{Z} = 0.6$ $P = \underline{\underline{9.598}}\ KW$ **Delta connected** $V_{Ph} = V_L = 400\ V$ $I_{Ph} = \frac{V_{Ph}}{Z} = \frac{400}{10} = \underline{\underline{40}}\ A$ Line current, $I_L = \sqrt{3}\, I_{Ph} = \underline{\underline{69.28}}\ A$ power absorbed, $P = \sqrt{3}\ V_L I_L \cos\phi$ $= \underline{\underline{28.8\ KW}}$